OpenClawBrain v12.2.6+: Shadow Routing with QTsim Confidence Mixing and Unified Policy Learning

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1 Abstract

OpenClawBrain is a TypeScript-first package workspace for retrieval routing in long-lived agents, deployed behind an OpenClaw-owned runtime boundary. The system is built around a strict hot/cold path split: the hot path is local and bounded for latency-critical routing, while the cold path runs asynchronous learning and structure updates. We define a *turn* as a single LLM API call. End-to-end cost is $C = \sum_{i=1}^{T} \text{tokens}_i \cdot \text{price}_i$, and the system targets lower T (fewer calls per user-visible request) and lower per-call token counts on the hot path. The system uses two route signals: a structural prior (graph_prior) and a query-time compatibility signal (QTsim). A confidence mixer combines those signals using uncertainty features, including entropy and margin, to decide how much weight each source gets per decision. Learning is performed through teacher distillation plus policy-gradient updates under a reward-source hierarchy. Human corrections have highest authority, then self-learning outcomes, then harvested signals, then high-volume teacher labels. This release includes RL-native maintenance of graph structure: Phase 2a is shipped, and the architecture defines connect/split/merge actions under policy control. The strongest current comparative artifacts are fixed-session benchmarks: `recorded_h2h_relational_drift_001` (800 queries, 41 steps, 8 baselines; `full_brain` 0.975 vs `vector_rag_rerank` 0.896) and sparse-feedback 10-seed evaluation (`full_brain` 91.96% vs best RAG 67.05% with ∼5x lower context/query).

2 1. System Overview

OpenClawBrain is designed around a strict split. OpenClawBrain owns memory structure and learning packages, while OpenClaw owns runtime orchestration and fail-open control:

- **Hot path:** local embedding retrieval and bounded traversal only.
- **Cold path:** asynchronous learning and structure updates.

We define a *turn* as a single LLM API call. Total cost can be written as

$$C = \sum_{i=1}^{T} \text{tokens}_i \cdot \text{price}_i$$

and OpenClawBrain reduces T and per-call token counts by keeping routing local and context bounded on the hot path, while allowing optional teacher calls only in the cold path.

1.1 Routing Inputs

At each route decision over candidate targets $\mathcal{A}(s)$ from state s:

- $\text{graph_prior}(a \mid s)$ captures persistent graph structure.
- $\text{QTsim}(a \mid q, s)$ captures query-conditioned fit.

1.2 Confidence Mixing

The router computes a query-local mix coefficient $\lambda \in [0, 1]$ from confidence features. Two required features are:

- **Entropy** of QTsim over candidates (high entropy means low confidence).
- **Margin** between top-1 and top-2 QTsim scores (low margin means ambiguity).

A simple form is:

$$\lambda = \sigma\big(\alpha_0 + \alpha_1 \, \text{margin}_{\text{QTsim}} - \alpha_2 \, H(\text{QTsim})\big)$$
$$\pi(a \mid q, s) = \lambda \, \text{QTsim}(a \mid q, s) + (1 - \lambda) \, \text{graph_prior}(a \mid s)$$

where σ is a logistic gate and H is entropy. When QTsim is sharp and separated, λ increases. When QTsim is uncertain, the policy falls back toward graph prior.

3 2. Learning Loop: Distillation + Policy Gradient

The learning loop combines distillation-style supervision with policy-gradient refinement.

2.1 Teacher Distillation

Offline replay samples route decisions and asks a teacher policy for preferred actions (or distributions). Distillation improves route quality quickly and provides dense supervision for states with sparse human labels.

2.2 Policy-Gradient Update

Given route trajectory $\tau = (s_0, a_0, \ldots, s_T, a_T)$ and scalar return $R(\tau)$:

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} (R(\tau) - b_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right]$$

where b_t is a baseline. Distillation targets and reward-weighted gradients are both applied; the exact blend is implementation-defined per run mode.

2.3 Reward-Source Hierarchy

The system resolves conflicting supervision by ordered authority:

1. Human correction/approval (highest authority)
2. Self-learning outcomes from deployed agents
3. Harvested weak signals
4. Teacher labels from async replay (lowest per-sample authority, highest volume)

This hierarchy constrains updates so high-authority signals can override lower-level noise.

4 3. RL-Native Maintenance (Structure Learning)

OpenClawBrain treats structure maintenance as a control problem over graph actions.

3.1 Phase 2a (Shipped)

Phase 2a ships RL-native maintenance controls that integrate policy outcomes with structure scoring and safe-apply gates. In practice, this means structure candidates are generated, scored with policy-derived signals, and applied only when guardrails pass.

3.2 Action Space: connect/split/merge

The policy-governed action space includes explicit graph edits:

- **Connect:** create/reweight candidate edges when repeated successful transitions lack direct structure.
- **Split:** divide overloaded nodes when policy confidence or error patterns indicate topic collapse.
- **Merge:** combine nodes with persistent redundancy and equivalent downstream behavior.

Each action class is executed with conservative thresholds, rollback support, and offline replay evaluation before broad enablement.

5 4. Evaluation: Evidence Ladder + Comparative Results

Evaluation is organized around a strict evidence ladder. The architecture split is fixed (OpenClaw runtime boundary, OpenClawBrain artifact boundary), and each claim is tied to a rung with its own artifact type.

4.1 Proof Boundary (What Is Proven vs Not Claimed)

- **Proven now:** deterministic mechanism proof plus fixed-session comparative benchmarks.
- **Not claimed in this paper:** shadow-traffic or online production superiority without those dedicated artifacts.
- **Rule:** benchmark wins are benchmark claims unless shadow/online artifacts are also published.

4.2 Top Comparative Rung: Recorded-Session H2H

Artifact: `recorded_h2h_relational_drift_001` (public bundle in `brain-ground-zero` at commit `1c54302`; see `docs/reproduce-eval.md`).

Interpretation: this rung demonstrates comparative advantage on a fixed recorded-session benchmark with shared state/query/rubric conditions.

Mode	Accuracy	Delta vs Best Baseline	Notes
`full_brain`	**0.975**	**+7.9 pp**	Best overall
`vector_rag_rerank`	0.89625	—	Best baseline

Table 1: Recorded-session head-to-head benchmark on a fixed relational-drift set (800 queries, 41 steps, 8 baselines).

Mode	Accuracy	Relative Context/Query	H2H Wins (10 seeds)
`full_brain`	**91.96%**	**1.0x**	**9**
Best RAG baseline	67.05%	∼5.0x	1

Table 2: Sparse-feedback 10-seed comparison. `full_brain` leads by +24.91 percentage points while using approximately 5x less context per query than the best RAG baseline.

4.3 Sparse-Feedback Benchmark (10 Seeds)

Sparse-feedback 10-seed results provide a second comparative rung focused on sparse supervision and context efficiency:

Interpretation: the learned full-brain policy remains stronger under sparse feedback while holding lower context/query cost.

4.4 Baselines and Ablations (Mechanism Layer)

We compare against three industry baselines plus the learned router in the reproducible harness:

- **Vector top-k** (`vector_only`): plain embedding retrieval, no traversal.
- **Top-k + reranker** (`edge_sim_legacy`): deterministic re-ranking with `route_mode=edge+sim`.
- **Pointer-chasing** (`edge_sim_legacy` with multi-hop traversal): iterative retrieval without learned routing.
- **OpenClawBrain learned** (`learned`) with ablation bounds `graph_prior_only` and `qtsim_only`.

These modes are exposed in the evaluation harness in the main `openclawbrain` repo.

4.5 Core Metrics

- **Reward / accuracy**: task success or correctness rate (dataset dependent).
- **% oracle gap closed**: fraction of the oracle gap closed over training epochs (expert-regions simulation).
- **Latency**: p50/p95 end-to-end query time from the eval harness.
- **Tool calls, tokens, cost**: measured in downstream agent runs; report as `TBD` if not available.

4.6 Figures (SVGs in `figures/eval`)

SVG sources live in `figures/eval/*.svg`; the PDF renders the PNG versions below.

4.7 Reporting Template (Do Not Invent Numbers)

All unreproduced results must be marked `TBD` and tied to an output file path.

Setting	Reward/Acc	% Oracle Gap	Latency p50/p95	Tool Calls	Tokens	Cost
Vector top-k (expert-regions sim)	0.878629 / 0.8920	TBD	TBD	TBD	TBD	TBD
Top-k + reranker	0.879762 / 0.8935	TBD	TBD	TBD	TBD	TBD
Pointer-chasing (graph prior only)	0.555147 / 0.5290	TBD	TBD	TBD	TBD	TBD
OpenClawBrain learned (mixed)	0.935126 / 0.9680	96.74%	TBD	TBD	TBD	TBD

Table 3: Expert-regions simulation baselines from reproducible runs. Each `TBD` must cite commit SHA, command, and artifact path (e.g., `/tmp/ocb_eval.json`).

4.8 Reproduction Workflow (TypeScript-First Package Surface)

Commands (from `openclawbrain` repo) and expected outputs:

```
cd /path/to/openclawbrain
corepack enable
pnpm install
pnpm check
pnpm release:pack

pnpm run eval:workflow-proof -- --output-dir scratch/workflow-proof/latest
pnpm run eval:baseline -- --queries /path/to/queries.jsonl --output /tmp/ocb_eval.json
pnpm run eval:sim-expert-regions -- --output-dir /tmp/ocb_expert_regions
pnpm run eval:sim-two-cluster -- --output-dir /tmp/ocb_two_cluster
```

Expected outputs include `/tmp/ocb_eval.json`, `/tmp/ocb_expert_regions/simulation_curve.csv`, and regenerated SVGs in `figures/eval/` using the plotting entrypoint documented in `docs/reproduce-eval.md`.

6 5. Operational Notes

OpenClawBrain supports local embeddings out-of-the-box for operator-first deployment. This enables deterministic bootstrap and private/offline-friendly operation while keeping optional teacher workflows in asynchronous jobs.

7 6. Conclusion

The v12.2.6+ system is defined by a fixed runtime boundary and four commitments: OpenClaw-owned runtime control, local hot-path retrieval, QTsim+graph-prior confidence mixing, distillation plus policy-gradient learning, and RL-native structure maintenance. The current top comparative rung is benchmark evidence (`recorded_h2h_relational_drift_001` and sparse-feedback 10-seed), while shadow/online claims remain explicitly out of scope until their artifacts are published.

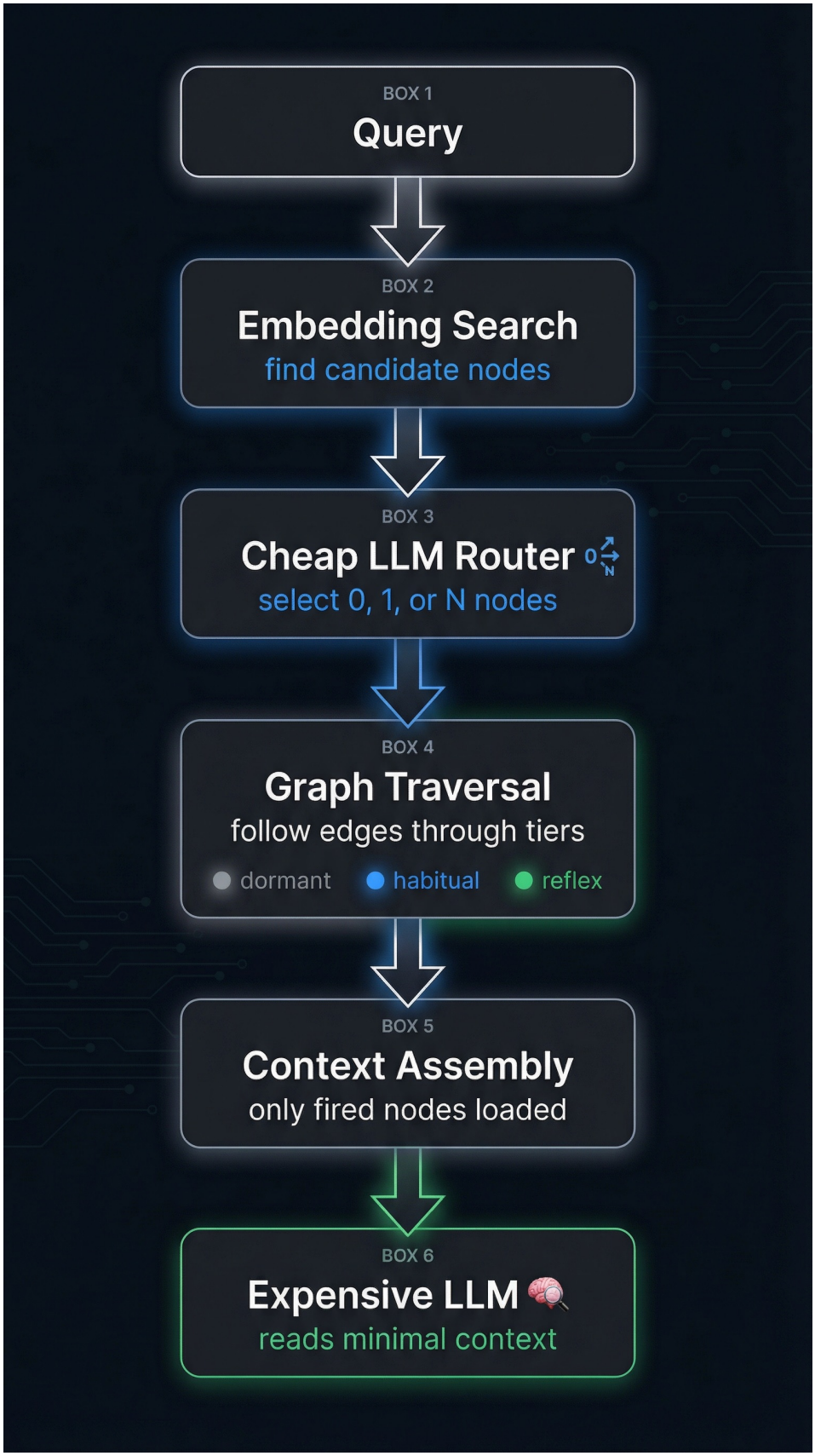

Figure 1: System diagram: latency-critical hot path stays local and bounded, while offline replay and policy updates run asynchronously in the cold path.

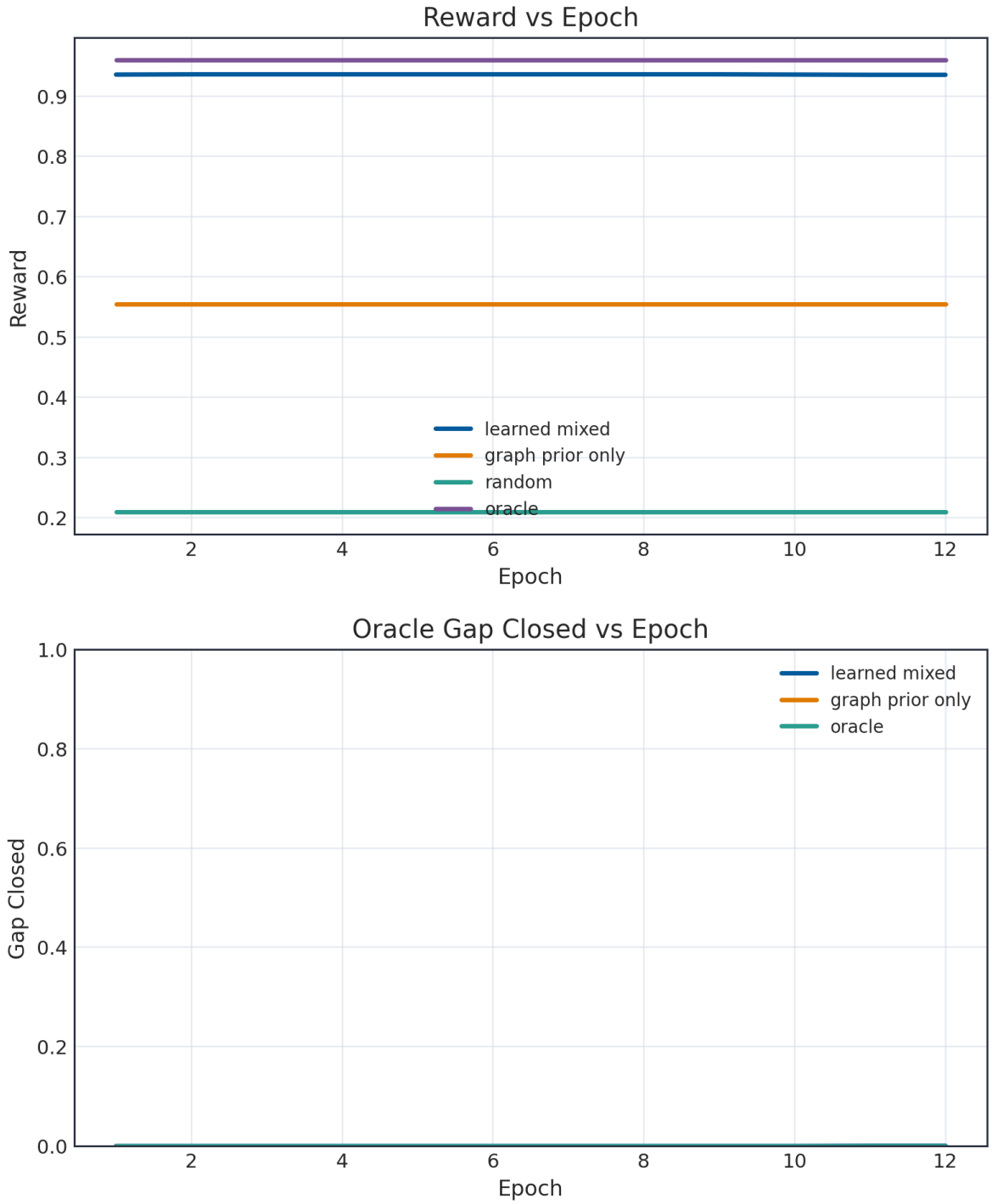

Figure 2: Expert-regions simulation learning behavior: reward trajectory and fraction of oracle gap closed across training epochs.

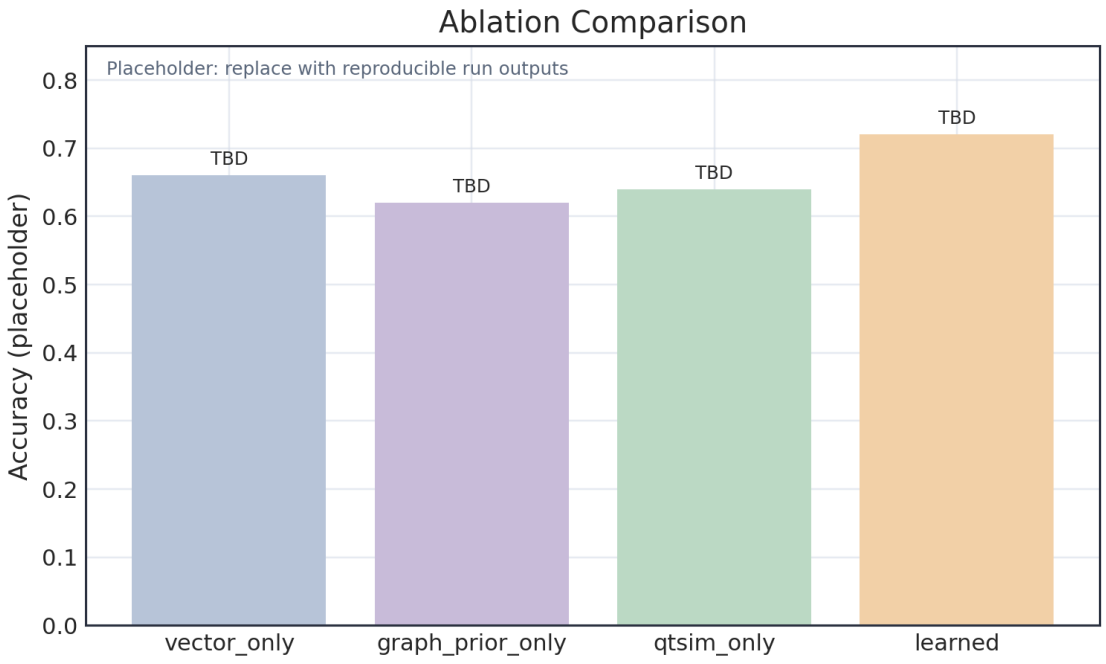

Figure 3: Ablation comparison across `vector_only`, `graph_prior_only`, `qtsim_only`, and learned routing.